The Eneström–Kakeya Theorem via Pandrosion Differences and Duality

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Abstract

We give a proof of the Eneström–Kakeya theorem using the "coefficient-space Pandrosion differences" $\Delta a_k = a_k - a_{k-1}$ and establish a *Pandrosion duality*: if P has increasing positive coefficients and all zeros in $|z| \leq 1$, then the Pandrosion quotient $Q(1, z) = (P(z) - P(1))/(z - 1)$ has decreasing positive (tail-sum) coefficients and all zeros in $|z| \geq 1$. We also show that the iterated Pandrosion quotients $Q_k(z)$ at $a = 0$, with coefficients $[a_k, a_{k+1}, \ldots, a_n]$, each satisfy the Eneström–Kakeya hypothesis, providing a hierarchy of zero-localization statements.

1 The Eneström–Kakeya theorem

Theorem 1.1 (Eneström 1893, Kakeya 1912)**.** *Let $P(z) = \sum_{k=0}^{n} a_k z^k$ with $0 < a_0 \leq a_1 \leq \cdots \leq a_n$. Then all zeros of P lie in $|z| \leq 1$.*

2 Proof via Pandrosion differences

Definition 2.1 (Coefficient Pandrosion differences)**.** The **Pandrosion differences** of the coefficient sequence $(a_0, a_1, \ldots, a_n)$ are:

$$\Delta a_k := a_k - a_{k-1}, \qquad k = 1, \ldots, n. \tag{1}$$

The Eneström–Kakeya condition is $\Delta a_k \geq 0$ for all k.

Proof of Theorem 1.1. **Step 1.** Expand $(z - 1) \cdot P(z)$:

$$\begin{aligned}
(z-1) \cdot P(z) &= \sum_{k=0}^{n} a_k z^{k+1} - \sum_{k=0}^{n} a_k z^k \\
&= a_n z^{n+1} - \sum_{k=1}^{n} (a_k - a_{k-1})\, z^k - a_0 \\
&= a_n z^{n+1} - \sum_{k=1}^{n} \Delta a_k \; z^k - a_0.
\end{aligned} \tag{2}$$

Step 2. For $|z| > 1$, apply the triangle inequality to (2):

$$\begin{aligned}
|(z-1)P(z)| &\geq a_n|z|^{n+1} - \sum_{k=1}^{n} \Delta a_k\,|z|^k - a_0 \\
&\geq a_n|z|^{n+1} - \left(\sum_{k=1}^{n} \Delta a_k\right)|z|^n - a_0 && \text{(since } |z|^k \leq |z|^n \text{ for } k \leq n\text{)} \\
&= a_n|z|^{n+1} - (a_n - a_0)\,|z|^n - a_0 && \text{(telescoping: } \textstyle\sum \Delta a_k = a_n - a_0\text{)} \\
&= a_n|z|^n(|z|-1) + a_0(|z|^n - 1) \\
&> 0 && \text{(since } |z| > 1,\ a_n > 0,\ a_0 > 0\text{).}
\end{aligned}$$

Step 3. Therefore $P(z) \neq 0$ for $|z| > 1$ (and $P(1) = \sum a_k > 0$), so all zeros lie in $|z| \leq 1$. □

Remark 2.2. The proof is classical (essentially Kakeya's original argument). The "Pandrosion" terminology emphasises that the *differences* Δa_k play the role of discrete divided differences in the coefficient space, analogous to the role of $Q(\alpha_k, z) = \prod_{j\neq k}(z-\alpha_j)$ in the root space. The telescoping $\sum \Delta a_k = a_n - a_0$ is the analogue of the Pandrosion sum identity $\sum_k Q(\alpha_k, z) = P'(z)$.

3 Pandrosion duality: P vs $Q(1,\cdot)$

Definition 3.1 (Tail-sum coefficients)**.** For a polynomial $P(z) = \sum_{k=0}^{n} a_k z^k$, the **tail sums** are:

$$b_j := \sum_{k=j+1}^{n} a_k, \qquad j = 0, 1, \ldots, n-1. \tag{3}$$

Theorem 3.2 (Pandrosion duality)**.** *Let P satisfy the Eneström–Kakeya hypothesis. Then:*

(i) *The Pandrosion quotient $Q(1,z) = (P(z) - P(1))/(z-1)$ is a polynomial of degree $n-1$ with coefficients $b_0, b_1, \ldots, b_{n-1}$ (the tail sums).*

(ii) *The tail sums are positive and* ***decreasing****: $b_0 \geq b_1 \geq \cdots \geq b_{n-1} = a_n > 0$.*

(iii) *All zeros of $Q(1,z)$ lie in $|z| \geq 1$.*

Proof. **Part (i).** Since $P(z) - P(1) = \sum_k a_k(z^k - 1) = (z-1)\sum_k a_k(z^{k-1} + \cdots + 1)$, dividing by $(z-1)$:

$$Q(1,z) = \sum_{k=1}^{n} a_k \sum_{j=0}^{k-1} z^j = \sum_{j=0}^{n-1}\left(\sum_{k=j+1}^{n} a_k\right) z^j = \sum_{j=0}^{n-1} b_j\, z^j.$$

Part (ii). $b_j - b_{j+1} = a_{j+1} \geq a_j > 0$ (using $a_{j+1} \geq a_j > 0$ from the E-K hypothesis). Note: the strict assertion $b_j > b_{j+1}$ holds because $a_{j+1} > 0$.

Part (iii). The *reciprocal polynomial* $z^{n-1}Q(1, 1/z) = \sum_{j=0}^{n-1} b_j z^{n-1-j} = \sum_{m=0}^{n-1} b_{n-1-m}\, z^m$ has **increasing** coefficients $b_{n-1} \leq b_{n-2} \leq \cdots \leq b_0$. By Theorem 1.1 applied to this reciprocal, its zeros lie in $|z| \leq 1$, hence the zeros of $Q(1,z)$ lie in $|z| \geq 1$. □

Remark 3.3 (The duality at a glance).

	$P(z)$	$Q(1,z)$
Coefficients	$a_0, a_1, \ldots, a_n$	$b_0, b_1, \ldots, b_{n-1}$
Ordering	increasing	decreasing
Zero region	$\lvert z\rvert \leq 1$	$\lvert z\rvert \geq 1$
Relation	$P(z) = (z-1)\cdot Q(1,z) + P(1)$	

The polynomial P and its Pandrosion quotient $Q(1,\cdot)$ are *dual*: P localises its zeros inside the unit disk, Q localises its zeros outside. The identity $P(z) = (z-1)Q(1,z) + P(1)$ links them arithmetically.

4 Iterated Pandrosion and the sub-polynomial hierarchy

Theorem 4.1 (Iterated Pandrosion at $a = 0$)**.** *Define $Q_0 = P$ and $Q_{m+1}(z) = (Q_m(z) - Q_m(0))/z$ for $m = 0, 1, \dots, n-1$. Then:*

(i) *$Q_m(z) = a_m + a_{m+1}z + \cdots + a_n z^{n-m}$, the "m-th truncation" of P.*

(ii) *Each Q_m satisfies the E-K hypothesis (since $a_m \le a_{m+1} \le \cdots \le a_n$).*

(iii) *All zeros of Q_m lie in $|z| \le 1$.*

(iv) *The Horner reconstruction $P(z) = a_0 + z \cdot Q_1(z) = a_0 + z(a_1 + z \cdot Q_2(z)) = \cdots$ expresses P as a nested composition of E-K polynomials.*

Proof. Part (i): $Q_0(z) = P(z)$, and $(Q_m(z) - Q_m(0))/z = (Q_m(z) - a_m)/z$, which removes the constant term and divides by z, shifting indices.

Parts (ii) and (iii): The subsequence $(a_m, a_{m+1}, \dots, a_n)$ inherits the E-K ordering. The conclusion follows from Theorem 1.1.

Part (iv): $P(z) = a_0 + z \cdot Q_1(z)$, and inductively $Q_m(z) = a_m + z \cdot Q_{m+1}(z)$. This is precisely Horner's evaluation scheme, reinterpreted as a Pandrosion decomposition. □

Remark 4.2. The Horner scheme is thus a *Pandrosion cascade*: each step peels off one coefficient by applying $Q(\cdot, 0)$, producing a sub-polynomial with the same E-K property. The zero set of P is "built up" from the zero sets of the sub-polynomials $Q_1, Q_2, \dots, Q_{n-1}$, all confined to the unit disk.

5 The Pandrosion annulus refinement

Proposition 5.1 (Inner radius bound)**.** *Under the E-K hypothesis, all zeros of P satisfy:*

$$\min_{0 \le k \le n-1} \frac{a_k}{a_{k+1}} \;\le\; |z| \;\le\; 1. \tag{4}$$

Proof. The upper bound is Theorem 1.1. For the lower bound, apply the "reverse E-K" argument to $z^n P(1/z) = a_n + a_{n-1}z + \cdots + a_0 z^n$, which has *decreasing* coefficients. By the E-K theorem for decreasing sequences (equivalent: all zeros of $z^n P(1/z)$ lie in $|z| \le \max_k (a_k/a_{k-1}) = 1/\min_k (a_k/a_{k+1})$), we get the lower bound.

An alternative proof uses Cauchy's bound: if $P(z_0) = 0$, then $|z_0| \ge a_0/(a_0 + |a_1| + \cdots + |a_n|) = a_0/P(1)$, but this is generally weaker. The sharper bound $\min a_k/a_{k+1}$ follows from Joyal–Labelle–Rahman (1967). □

6 Conclusion

The Eneström–Kakeya theorem admits a natural formulation in Pandrosion language:

Classical	**Pandrosion**
$a_k - a_{k-1} \geq 0$	Pandrosion differences $\Delta a_k \geq 0$
$(z-1)P(z) = a_n z^{n+1} - \sum \Delta a_k z^k - a_0$	Pandrosion factorisation
$Q(1, z)$ coefficients	Tail sums $b_j = \sum_{k>j} a_k$
P zeros in $\lvert z\rvert \leq 1$	"Internal" localisation
$Q(1, \cdot)$ zeros in $\lvert z\rvert \geq 1$	"External" localisation
Horner scheme	Pandrosion cascade at $a = 0$

The main new contributions are:

1. The **Pandrosion duality** (Theorem 3.2): P and $Q(1, \cdot)$ have complementary zero regions ($|z| \leq 1$ vs $|z| \geq 1$) and complementary coefficient orderings (increasing vs decreasing).
2. The **iterated Pandrosion hierarchy** (Theorem 4.1): the Horner scheme is a cascade of E-K polynomials, each with zeros in the unit disk.

References

[1] G. Eneström, *Remarque sur un théorème relatif aux racines de l'équation $a_n x^n + \cdots + a_0 = 0$ où tous les coefficients sont réels et positifs*, Tôhoku Math. J. **18** (1920), 34–36. (Original: Öfversigt af Vetenskaps-Akademiens Förhandlingar **50**, 1893.)

[2] S. Kakeya, *On the limits of the roots of an algebraic equation with positive coefficients*, Tôhoku Math. J. **2** (1912), 140–142.

[3] A. Joyal, G. Labelle, Q.I. Rahman, *On the location of zeros of polynomials*, Canad. Math. Bull. **10** (1967), 53–63.

[4] I. Besevic, *Pandrosion's cube duplication and derivative-free iterations*, Preprint (2026).